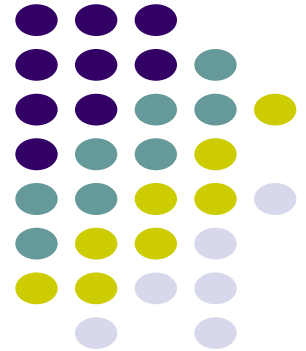


RAID

ECE 469, April 23

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RAID



- Two motivations
 - (in the past) Operating in parallel can increase disk throughput
 - RAID = Redundant Array of Inexpensive Disks
 - (today) Redundancy can increase reliability
 - RAID = Redundant Array of Independent Disks

RAID

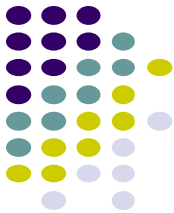


- Parallel reading (striping) (for performance)
 - Splitting bits of a byte across multiple disks
 - 8 disks (bit-level striping)
 - Logically acts like single disk with sector size * 8 and access time / 8
 - Reduce the response time of large access (e.g. one 4K block)
 - Alternatively, block-level striping
 - Increases the throughput of multiple small accesses (e.g. eight 512-byte blocks)

RAID



- Mirroring or shadowing ([for reliability](#))
 - Local disk consists of 2 physical disks in parallel
 - Every write performed on both disks
 - Can read from either disk
 - Probability of both fail at the same time?



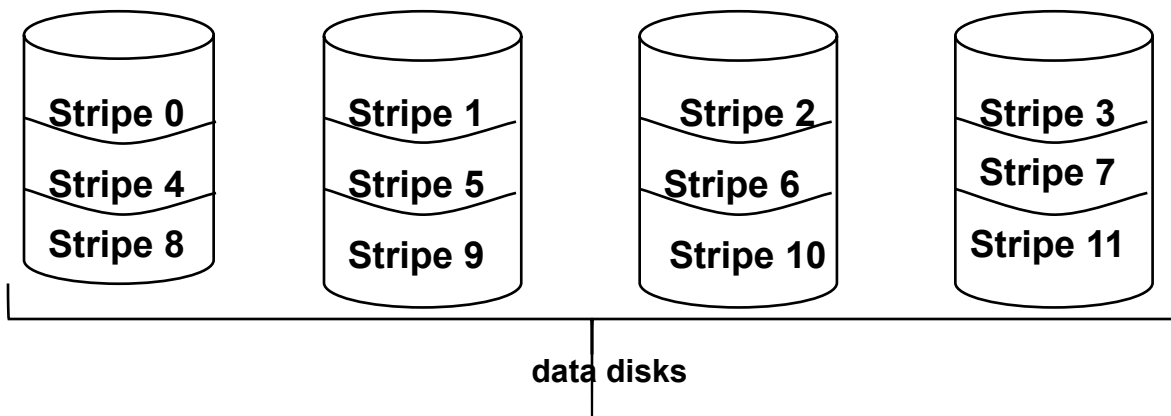
RAID - Combine the two ideas

- Mirroring gives reliability, but expensive
- Striping gives high data-transfer rate, but not reliability
- Challenge: can we provide redundancy at low cost?



RAID Level 0

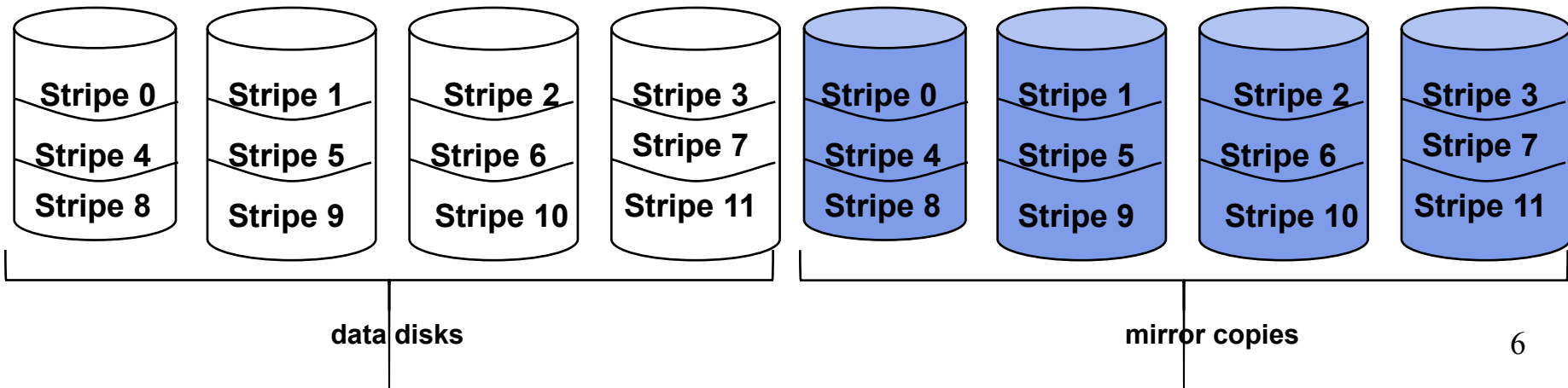
- Level 0 is non-redundant disk array
- Files are Striped (**block level**) across disks, no redundant info
- High read throughput
- Best write throughput among RAID levels (no redundant info to write)
- Any disk failure results in data loss
 - What's the MTTF (mean time to failure) of the whole system?





RAID Level 1

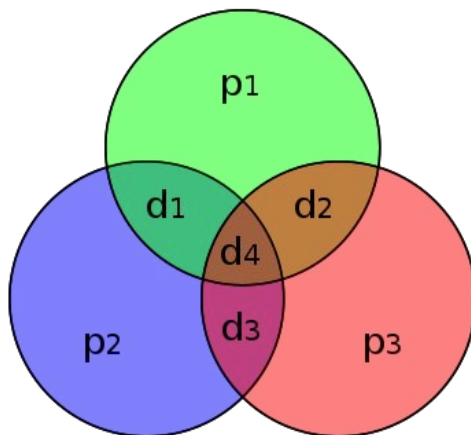
- Mirrored Disks
- Data is written to two places
 - On failure, just use surviving disk
- On read, choose fastest to read
 - Write performance is same as single drive, read performance is 2x better
- Expensive





Error Correcting Code : Hamming Code

- Can we recover from errors without creating complete back up?
- Error Correcting Code:
 - E.g., Hamming Code
 - Can detect multiple bit-errors and can fix single bit error.

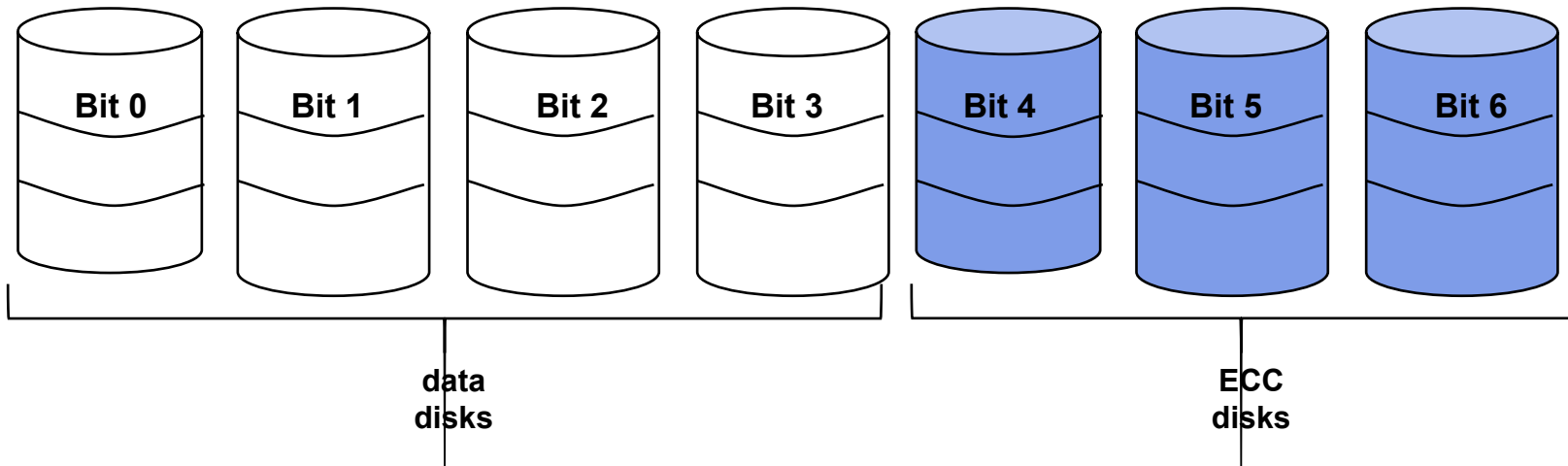


- p => Parity Bits
- d => Data bits



RAID Level 2

- Bit-level Striping with Hamming (ECC) codes for error correction
- All 7 disk arms are synchronized and move in unison
- Complicated controller
- Single access at a time
- Tolerates only one error, but with no performance degradation
- Not used in real world





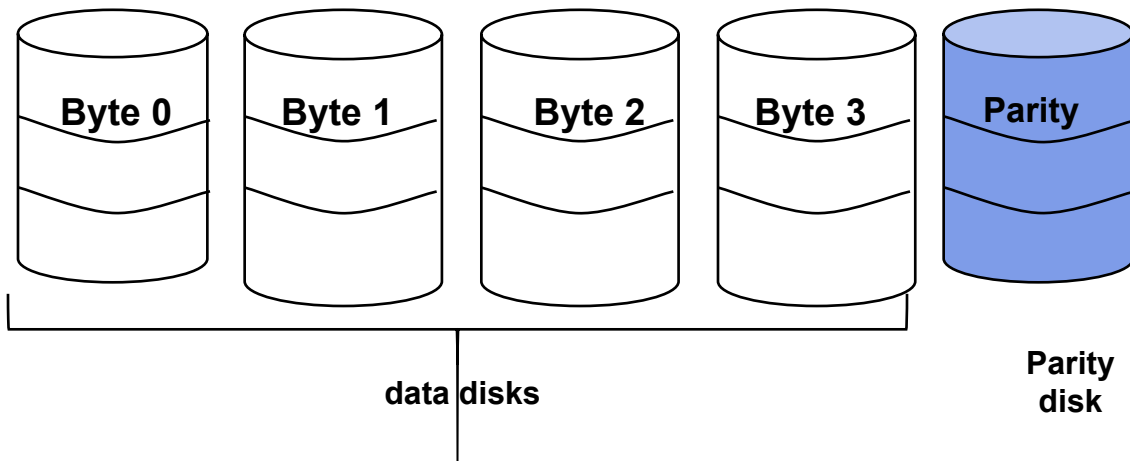
Parity Bits

- What do you need to do in order to detect and correct a one-bit error ?
 - Suppose you have a binary number, represented as a collection of bits: $\langle b_3, b_2, b_1, b_0 \rangle$, e.g. 0110
- Detection is easy
- Parity:
 - Count the number of bits that are on, see if it's odd or even
 - EVEN parity is 0 if the number of 1 bits is even
 - $\text{Parity}(\langle b_3, b_2, b_1, b_0 \rangle) = P_0 = b_0 \otimes b_1 \otimes b_2 \otimes b_3$
 - $\text{Parity}(\langle b_3, b_2, b_1, b_0, p_0 \rangle) = 0$ if all bits are intact
 - $\text{Parity}(0110) = 0$, $\text{Parity}(01100) = 0$
 - $\text{Parity}(11100) = 1 \Rightarrow \text{ERROR!}$
 - Parity can detect a single bit error, but can't tell you which of the bits got flipped



RAID Level 3

- Use a parity disk
 - Each byte on the parity disk is a parity function of the corresponding bytes on all the other disks
- A read accesses all the data disks
- A write accesses all data disks plus the parity disk
- **On disk failure, read remaining disks plus parity disk to compute the missing data**

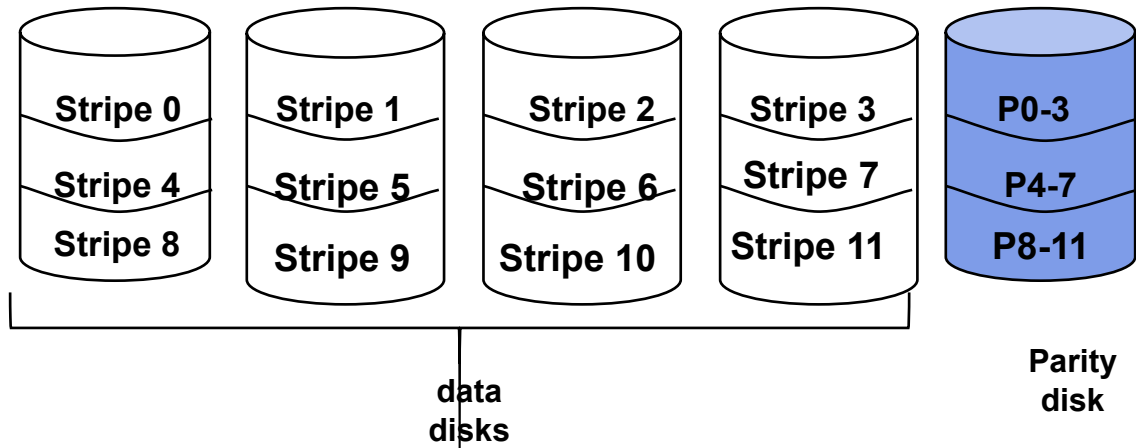


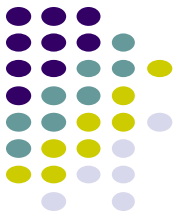
Single parity disk can be used to detect and recover errors



RAID Level 4

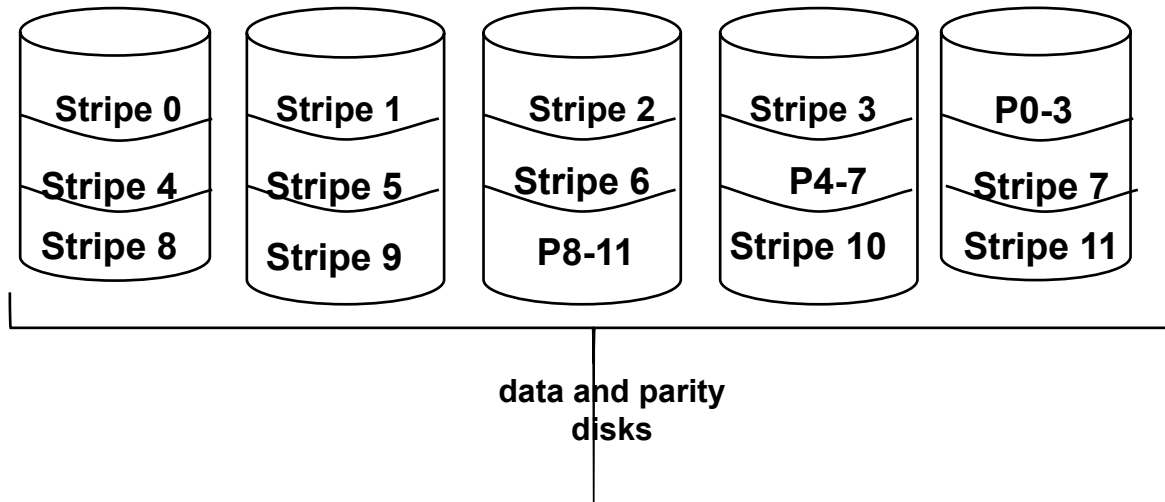
- Combines Level 0 and 3 – block-level parity with Stripes
- Lower transfer rate for each block (by single disk)
- Higher overall rate (many small files, or a large file)
- Large writes □ parity bits can be written in parallel
- Small writes □ 2 reads + 2 writes !
- **Heavy load on the parity disk**

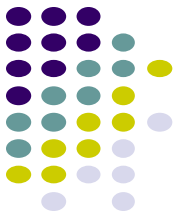




RAID Level 5

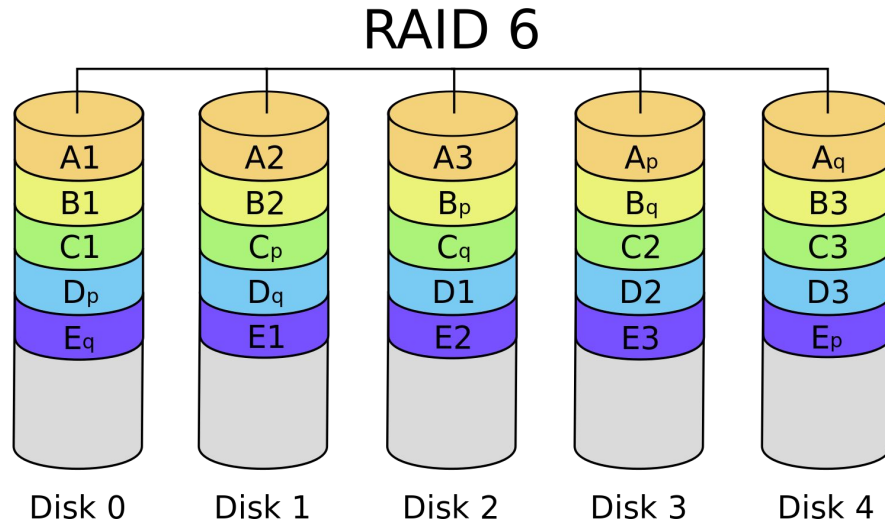
- Block Interleaved Distributed Parity
- Like parity scheme, but distribute the parity info over all disks (as well as data over all disks)
- Better (large) write performance
 - **No single disk as performance bottleneck**





RAID Level 6

- Level 5 with an extra parity bit
- **Can tolerate two failures**
 - What are the odds of having two concurrent failures ?
- No performance penalty on reads, slower on writes (compared to RAID5)

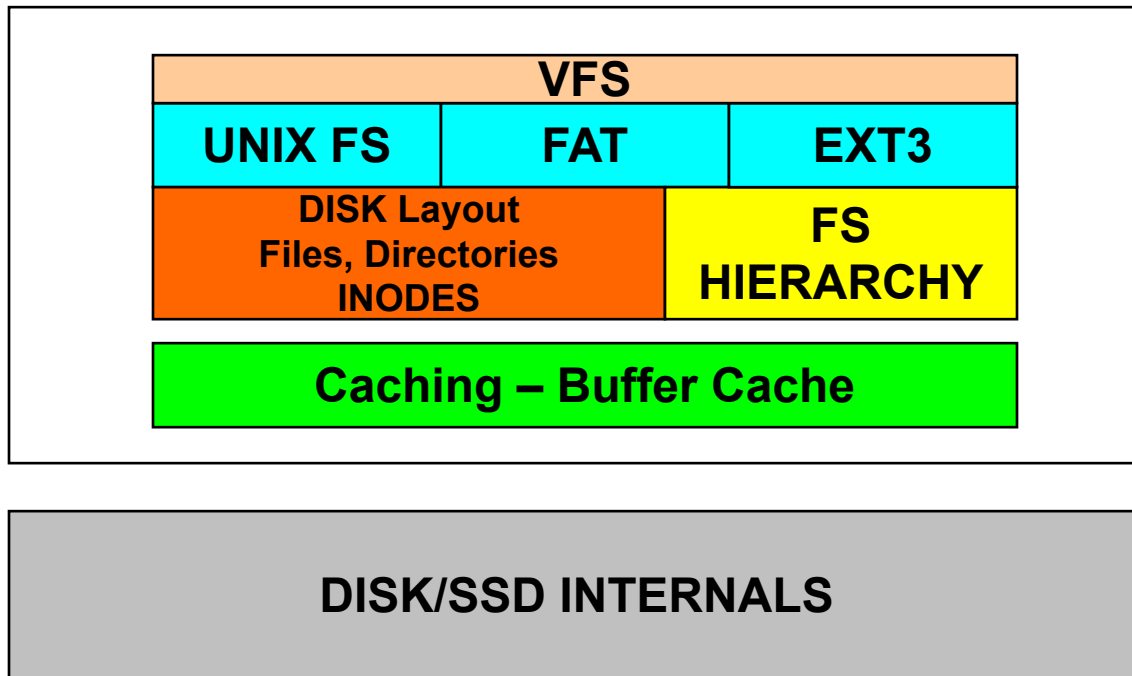




RAID Implementation

- Typically in hardware
 - Special-purpose RAID controller (PCI card)
 - Manages disks
 - Performs parity calculation
- Can be in software (by OS)
 - Can be fast
 - At the cost of CPU time

FS Topics Covered Till now!



**FS
Reliability**

**Crash
Recovery**

Journaling

RAID